

☒ Easy Level (10 Questions)

1.

Which software allows you to create and run virtual machines on a desktop?

- A) GCC
- B) VirtualBox
- C) bash
- D) Apache

Answer: B

☒ VirtualBox is a Type-2 hypervisor.

2.

Which type of hypervisor runs directly on physical hardware?

- A) Type 2
- B) Hosted
- C) Bare-metal
- D) VirtualBox

Answer: C

☒ Bare-metal hypervisors like KVM and ESXi run directly on hardware.

3.

What does a snapshot do in VM management?

- A) Deletes the VM
- B) Installs new OS
- C) Captures the current VM state
- D) Clones the disk

Answer: C

4.

Which command-line tool manages KVM-based VMs?

- A) top
- B) virsh
- C) nano
- D) docker

Answer: B

5.

Which network mode allows VM to access the internet via the host?

- A) Internal
- B) NAT
- C) Host-Only
- D) Loopback

Answer: B

6.

Which format is used for VirtualBox disk files?

- A) .vdi
- B) .img
- C) .qcow
- D) .ova

Answer: A

7.

What is a clone in VM terms?

- A) Snapshot
- B) Backup
- C) Copy of an existing VM
- D) Log file

Answer: C

8.

Which mode allows VM to act like a real device on the LAN?

- A) NAT
- B) Bridged
- C) Host-only
- D) Tunnel

Answer: B

9.

Which command shows IP information in Linux?

- A) ps
- B) ping
- C) ip a
- D) vi

Answer: C

10.

Which file stores the configuration of a libvirt VM?

- A) .bashrc
- B) /etc/passwd
- C) XML file
- D) .vimrc

Answer: C

☒ Medium Level (20 Questions)

11.

Which tool is used to create virtual networks in KVM?

- A) brctl
- B) virsh net-*
- C) VBoxManage
- D) nmcli only

Answer: B

12.

Which of the following is a benefit of using snapshots?

- A) Security
- B) Monitoring
- C) Quick rollback
- D) Bandwidth optimization

Answer: C

13.

What is the purpose of MAC address filtering in VMs?

- A) Increase speed
- B) Assign hostname

- C) Enhance security
- D) Allocate memory

Answer: C

14.

Which command lists all active VMs in virsh?

- A) virsh status
- B) virsh list --all
- C) virsh show
- D) vmstat

Answer: B

15.

Which of the following provides direct device access in virtual networks?

- A) NAT
- B) SR-IOV
- C) VLAN
- D) MAC bridge

Answer: B

16.

What does a "host-only" network allow?

- A) Access to the internet
- B) Only communication between VM and host
- C) Communication across LAN
- D) Public access

Answer: B

17.

Which networking mode is best for isolating VMs from outside access?

- A) Bridged
- B) NAT
- C) Internal
- D) Host-Only

Answer: C

18.

Which component of KVM/QEMU acts as the virtual switch?

- A) libvirt
- B) virbr0
- C) ifconfig
- D) nmap

Answer: B

19.

What is a virtual NIC?

- A) Physical network interface
- B) Logical network card inside VM
- C) Load balancer
- D) USB storage

Answer: B

20.

Which hypervisor is part of Linux kernel?

- A) VirtualBox
- B) Hyper-V
- C) KVM
- D) VMware Player

Answer: C

21.

What is the main purpose of VLANs in virtual networks?

- A) Port forwarding
- B) Encrypt data
- C) Traffic segmentation
- D) Compress logs

Answer: C

22.

What is the role of VBoxManage?

- A) Configure Apache
- B) Install packages
- C) Manage VirtualBox VMs
- D) Backup MySQL

Answer: C

23.

Which format is used by KVM for virtual disk images?

- A) .qcow2
- B) .vmdk
- C) .ova
- D) .iso

Answer: A

24.

Which command configures network interface IP manually in Linux?

- A) ip addr add
- B) ping
- C) netstat
- D) ifup

Answer: A

25.

Which command shuts down a running VM in virsh?

- A) virsh poweroff
- B) virsh destroy
- C) virsh off
- D) virsh pause

Answer: B

☒ It immediately powers off (like unplugging), while shutdown does graceful.

26.

What is one risk of using bridged networking?

- A) VM runs slowly

- B) No internet
- C) VM exposed to LAN attacks
- D) Host firewall disabled

Answer: C

27.

Which tool helps manage virtual NICs and bridges in Linux?

- A) iftop
- B) ip
- C) nmcli
- D) Both B and C

Answer: D

28.

What file type is used to export a full VM for sharing?

- A) .vmdk
- B) .ova
- C) .bin
- D) .dd

Answer: B

29.

How to create a new virtual bridge in KVM with virsh?

- A) virsh net-create
- B) virsh bridge-new
- C) ip link create
- D) brctl addbr

Answer: A

30.

What is the advantage of SR-IOV?

- A) Virtual snapshots
- B) Disk compression
- C) Direct NIC hardware access from VM
- D) NAT translation

Answer: C

☒ Hard Level (10 Questions)

31.

Which layer does bridged networking operate at?

- A) Application Layer

- B) Transport Layer
- C) Network Layer
- D) Data Link Layer

Answer: D

32.

Which kernel module is required to enable KVM virtualization?

- A) kvm_intel / kvm_amd
- B) qemu-kvm
- C) libvirt-daemon
- D) bridge-utils

Answer: A

33.

What happens when a snapshot is reverted?

- A) The VM is updated
- B) VM goes to sleep
- C) All current changes are discarded
- D) Only RAM is cleared

Answer: C

34.

Which technique enables dynamic migration of running VMs?

- A) SR-IOV
- B) Live migration
- C) Cold migration
- D) IP switching

Answer: B

35.

Which file defines VM settings in libvirt?

- A) .vminfo
- B) .libxml
- C) domain.xml
- D) vm_config.conf

Answer: C

36.

What is the danger of using shared bridges for all VMs?

- A) Slower boot
- B) Network collisions & sniffing
- C) Memory leaks
- D) Kernel panic

Answer: B

37.

Which command creates a virtual disk in VirtualBox?

- A) vboxdisk create
- B) VBoxManage createhd
- C) virsh diskcreate
- D) vboxinit

Answer: B

38.

What does virsh net-autostart do?

- A) Enables bridging
- B) Deletes default network
- C) Sets network to start with host boot
- D) Removes DHCP

Answer: C

39.

Which config in KVM enables VM access to the physical network?

- A) virbr0
- B) br0 bridged to eth0
- C) lo
- D) tap device

Answer: B

40.

What is the maximum number of vNICs supported in most hypervisors?

- A) 1
- B) 4
- C) 8
- D) Depends on hypervisor

Answer: D